

MemComputing in the age of AI

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MemComputing is a new physics-based approach to computation that employs time non-locality (memory) to both process and store information on the same physical location. In this talk, I will discuss its application in the field of Machine Learning, by showing efficient supervised and unsupervised training of neural networks, and demonstrating its advantages over traditional sampling methods.

Index Terms—MemComputing, Machine Learning, Artificial Intelligence

I. INTRODUCTION

MEMCOMPUTING is a computing paradigm that employs memory (time non-locality) to process information [1]–[4]. MemComputing machines can operate either in the analog or the digital regime [5]. Digital MemComputing machines (DMMs) are of particular interest since they are easily scalable and can be directly applied to a wide variety of combinatorial optimization problems [3]. It was indeed shown that DMMs demonstrate considerable advantages in the solution of these problems compared to traditional algorithms [6]–[9].

II. ACCELERATING NEURAL NETWORK TRAINING

Training of artificial neural networks (NNs) belongs to this class of problems. It is a particularly difficult task as it requires matching the probability distribution of the input data with a given statistical model. In turn, this requires navigating a non-convex landscape to find the mode(s) of such a distribution or a very good approximation to them. In this talk, I will show several applications of DMMs in the efficient training of NNs in both the supervised [10] and unsupervised setting [11], [12]. I will show that DMMs facilitate the training of NNs at optimality and as a single unit, thus requiring orders of magnitude smaller number of nodes. In the hardware realization of these machines, this gain would directly translate into a much more energy efficient training of NNs, and consequently more energy-efficient inference. These advantages transfer also to certain types of quantum problems [9].

III. IMPROVING DMMs WITH AI

In turn, I will show how to use Artificial Intelligence (AI) to improve on the design of these machines. In particular, I will show that scientific discovery can be formulated as meta-optimization using large language models (LLMs) as agents for such a discovery [9]. In this framework, the optimization objective itself is also being optimized, using a “consensus objective aggregation,” where LLM-generated objective functions are combined via correlation-weighted voting, yielding a stable, self-correcting evaluation criterion that evolves as understanding deepens. We have applied such a framework to algorithm discovery for 3-SAT problems based on DMMs and have been able to reduce the baseline scaling exponent by 50% with about 67x speedup on the largest instances tested [13].

A. Conclusion

In conclusion, I will show that MemComputing can both help in the field of AI and, vice versa, AI can be useful in improving on the design of these machines.

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